

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE:CSA16 03

COURSE NAME: Data Warehousing and Data Mining

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COURSE OUTCOME COVERED

CO4: Examine the applicability of clustering algorithms in real-world scenarios, presenting findings through clear reports with effective visualizations.

Assessment Tool Weightage: 40%

QUESTIONS: 05

Total Marks:50

Annexure A

Industry Problem Solving Task

Criteria	Understanding of Industry Problem (2)	Application of Engineering Concepts (2.5)	Solution Design (2.5)	Use of Tools (2)	Analysis (1)	Total(10)
Weightage	20%	25%	25%	20%	10%	100%
<i>Q1</i>						
<i>Q2</i>						
<i>Q3</i>						
<i>Q4</i>						
<i>Q5</i>						

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

Annexure A

Q1 . Dataset: Supermarket Purchases

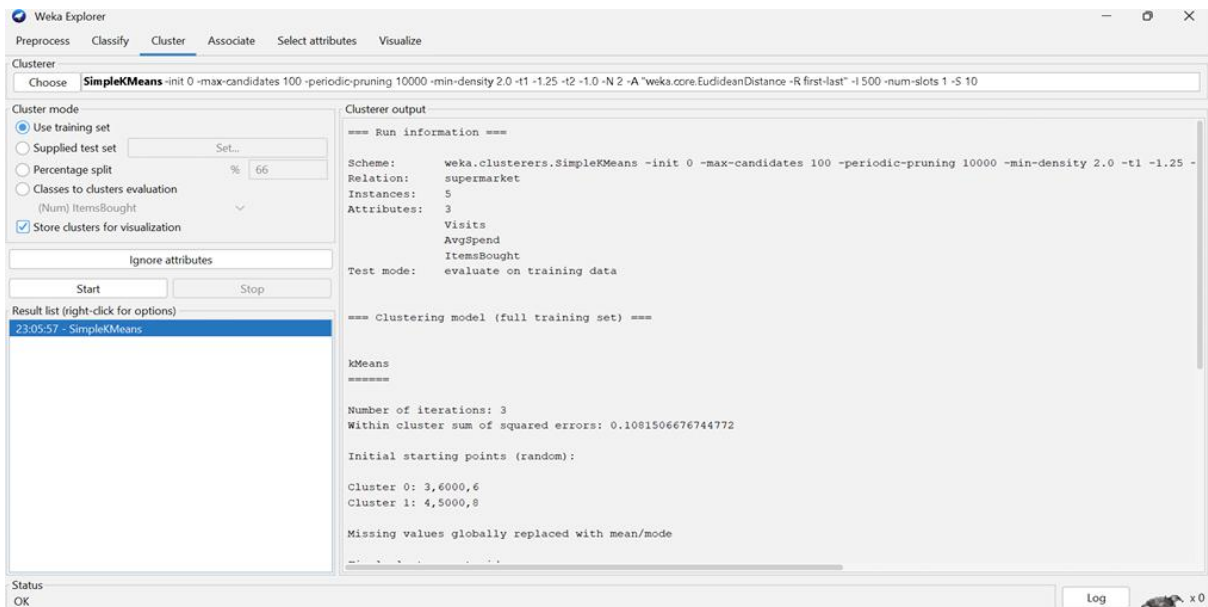
Customer Visits/Month Avg Spend (₹) Items Bought

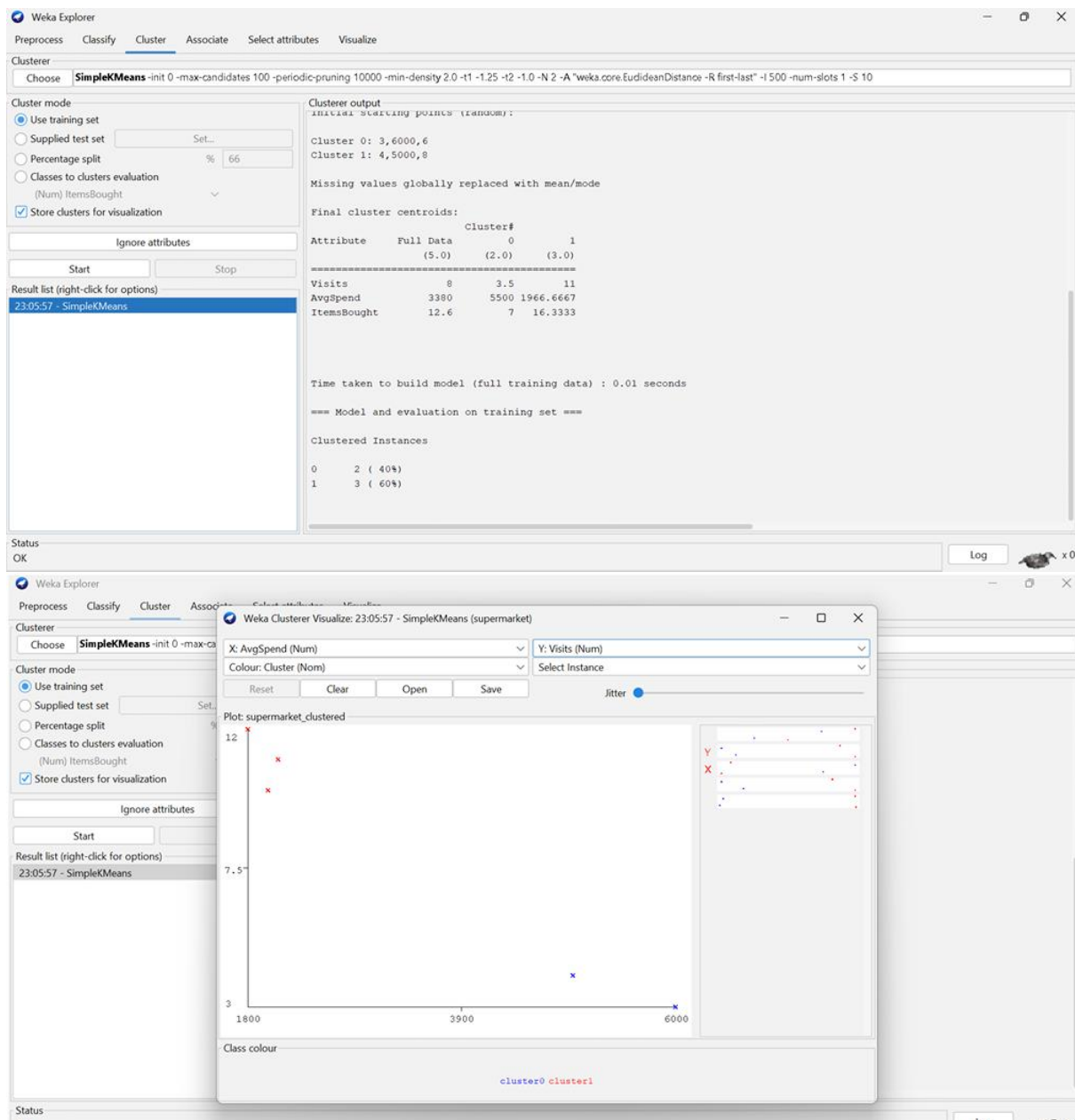
C1	10	2000	15
C2	4	5000	8
C3	12	1800	18
C4	3	6000	6
C5	11	2100	16

Question:

Using the above dataset, propose a clustering approach to segment customers and explain how it supports targeted marketing. (BL3 – 7 Marks)

ANS:





Q2. Dataset: Credit Card Transactions

Transaction Amount (₹) Location Time (hrs)

T1	200	Chennai	10
T2	5000	Delhi	2
T3	150	Chennai	11
T4	7000	Mumbai	1
T5	180	Chennai	12

Question:

Design a clustering-based solution using this dataset to detect unusual or fraudulent transactions. (BL4 – 7 Marks)

ANS

Weka Explorer

Preprocess **Cluster** Associate Select attributes Visualize

Clusterer: Choose **SimpleKMeans** -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 2 -A "weka.core.EuclideanDistance -R first-last" -I 500 -num-slots 1 -S 10

Cluster mode:
☒ Use training set
☐ Supplied test set Set...
☐ Percentage split % 66
☐ Classes to clusters evaluation (Num) Time
☒ Store clusters for visualization

Ignore attributes: _____

Start Stop

Result list (right-click for options):
 23:05:57 - SimpleKMeans
 23:11:41 - SimpleKMeans

Status: OK

Cluster output:

```

=== Run information ===

Scheme:      weka.clusterers.SimpleKMeans -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 2 -A "weka.core.EuclideanDistance -R first-last" -I 500 -num-slots 1 -S 10
Relation:    creditcard
Instances:    5
Attributes:   3
              Amount
              LocationCode
              Time
Test mode:    evaluate on training data

=== Clustering model (full training set) ===

KMeans
=====

Number of iterations: 3
Within cluster sum of squared errors: 0.1683116267710061

Initial starting points (random):

Cluster 0: 7000,2,1
Cluster 1: 5000,1,2

Missing values globally replaced with mean/mode
  
```

Log

Weka Explorer

Preprocess **Cluster** Associate Select attributes Visualize

Clusterer: Choose **SimpleKMeans** -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 2 -A "weka.core.EuclideanDistance -R first-last" -I 500 -num-slots 1 -S 10

Cluster mode:
☒ Use training set
☐ Supplied test set Set...
☐ Percentage split % 66
☐ Classes to clusters evaluation (Num) Time
☒ Store clusters for visualization

Ignore attributes: _____

Start Stop

Result list (right-click for options):
 23:05:57 - SimpleKMeans
 23:11:41 - SimpleKMeans

Status: OK

Cluster output:

```

Initial starting points (random):

Cluster 0: 7000,2,1
Cluster 1: 5000,1,2

Missing values globally replaced with mean/mode

Final cluster centroids:

Attribute      Cluster#
              (5.0) (2.0) (3.0)
=====
Amount         2506  6000  176.6667
LocationCode    0.6   1.5    0
Time           7.2   1.5   11

Time taken to build model (full training data) : 0 seconds

=== Model and evaluation on training set ===

Clustered Instances

0      2 ( 40%)
1      3 ( 60%)
  
```

Log

Weka Explorer

Preprocess **Cluster** Associate Select attributes Visualize

Clusterer: Choose **SimpleKMeans** -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 2 -A "weka.core.EuclideanDistance -R first-last" -I 500 -num-slots 1 -S 10

Cluster mode:
☒ Use training set
☐ Supplied test set Set...
☐ Percentage split % 66
☐ Classes to clusters evaluation (Num) Time
☒ Store clusters for visualization

Ignore attributes: _____

Start Stop

Result list (right-click for options):
 23:05:57 - SimpleKMeans
 23:11:41 - SimpleKMeans

Status: OK

Weka Clusterer Visualizer: 23:11:41 - SimpleKMeans (creditcard)

X: Instance_number (Num) Y: Amount (Num)
 Colour: Cluster (Nom) Select Instance

Reset Clear Open Save Jitter

Plot: creditcard_clustered

Class colour: cluster0 cluster1

Q3. Dataset: Telecom Usage

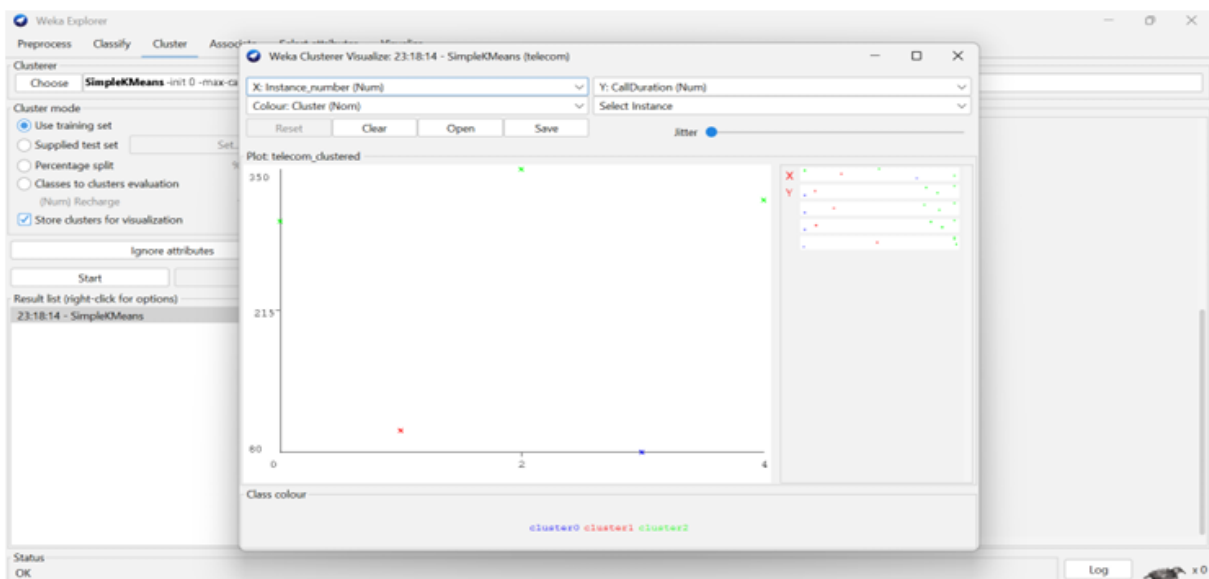
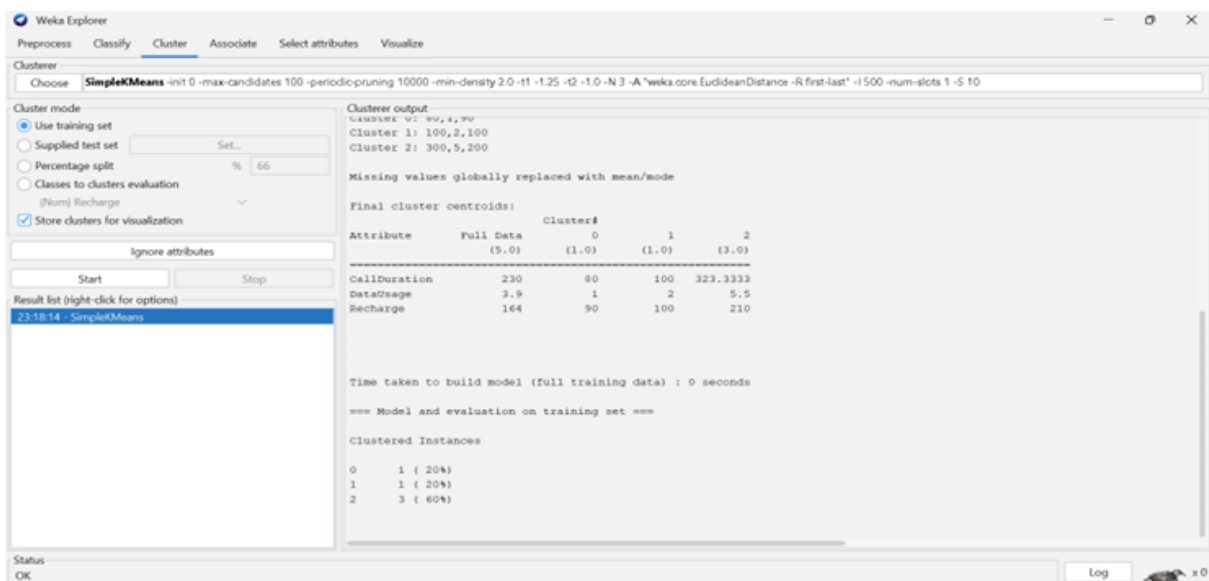
User Call Duration (min) Data Usage (GB) Recharge (₹)

U1	300	5	200
U2	100	2	100
U3	350	6	220
U4	80	1	90
U5	320	5.5	210

Question:

Explain how clustering can be applied to identify churn-prone users and improve customer retention strategies. (BL4 – 7 Marks)

ANS



Q4 . Dataset: E-commerce Behavior

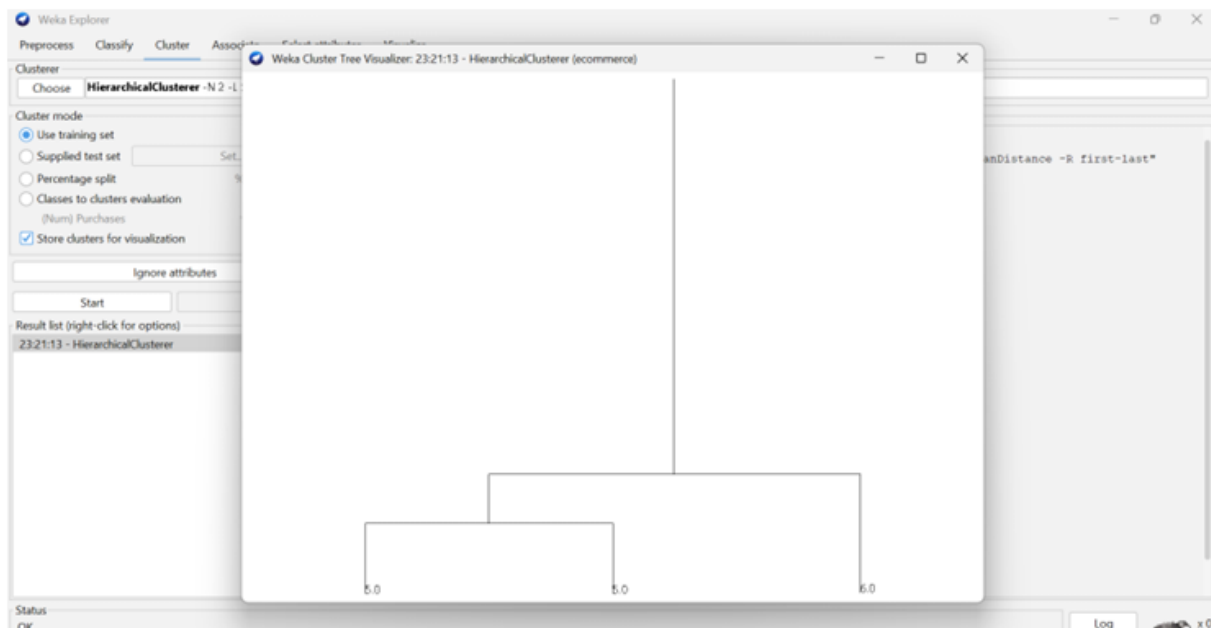
User Product Views Clicks Purchases

U1	50	20	5
U2	30	10	2
U3	60	25	6
U4	20	8	1
U5	55	22	5

Question:

Suggest a clustering technique to group users and explain how it improves recommendation systems. (BL3 – 7 Marks)

ANS



Q5. Dataset: Traffic Data

Location Vehicle Count Avg Speed (km/h) Delay (min)

L1	200	30	10
L2	500	15	25
L3	220	28	12
L4	600	10	30
L5	210	32	9

Question:

Design a clustering-based solution to identify traffic congestion zones and explain how it helps in traffic management. (BL4 – 7 Marks)

ANS

Preprocess Classify **Cluster** Associate Select attributes Visualize

Clusterer

Choose **SimpleKMeans** -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 2 -A "weka.core.EuclideanDistance -R first-last" -i500 -num-slots 1 -S 10

Cluster mode

☒ Use training set

☐ Supplied test set

☐ Percentage split

☐ Classes to clusters evaluation

☐ (Num) Delay

☒ Store clusters for visualization

Ignore attributes

Start Stop

Result list (right-click for options)

23:24:05 - SimpleKMeans

Cluster output

--- Run information ---

Scheme: weka.clusterers.SimpleKMeans -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25

Relation: traffic

Instances: 5

Attributes: 3

VehicleCount

AvgSpeed

Delay

Test mode: evaluate on training data

--- Clustering model (full training set) ---

KMeans

Number of iterations: 3

Within cluster sum of squared errors: 0.11378205368465107

Initial starting points (random):

Cluster 0: 600,10,30

Cluster 1: 500,15,25

Missing values globally replaced with mean/mode

Log

Weka Explorer

Preprocess Classify **Cluster** Associate Select attributes Visualize

Clusterer

Choose **SimpleKMeans** -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 2 -A "weka.core.EuclideanDistance -R first-last" -i500 -num-slots 1 -S 10

Cluster mode

☒ Use training set

☐ Supplied test set

☐ Percentage split

☐ Classes to clusters evaluation

☐ (Num) Delay

☒ Store clusters for visualization

Ignore attributes

Start Stop

Result list (right-click for options)

23:24:05 - SimpleKMeans

Cluster output

ANALYZE STARTING POINTS (RANDOM):

Cluster 0: 600,10,30

Cluster 1: 500,15,25

Missing values globally replaced with mean/mode

Final cluster centroids:

Attribute	Full Data	Cluster# 0	Cluster# 1
	(5.0)	(2.0)	(3.0)
VehicleCount	346	550	210
AvgSpeed	23	12.5	30
Delay	17.2	27.5	10.3333

Time taken to build model (full training data) : 0 seconds

--- Model and evaluation on training set ---

Clustered Instances

Cluster	Count	Percentage
0	2	40%
1	3	60%

Weka Explorer

Preprocess Classify **Cluster** Associate Select attributes Visualize

Clusterer

Choose **SimpleKMeans** -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 2 -A "weka.core.EuclideanDistance -R first-last" -i500 -num-slots 1 -S 10

Cluster mode

☒ Use training set

☐ Supplied test set

☐ Percentage split

☐ Classes to clusters evaluation

☐ (Num) Delay

☒ Store clusters for visualization

Ignore attributes

Start Stop

Result list (right-click for options)

23:24:05 - SimpleKMeans

Weka Cluster Visualizer: 23:24:05 - SimpleKMeans (traffic)

X: Instance_number (Num)

Y: VehicleCount (Num)

Colour: Cluster (Nom)

Select Instance

Plot traffic clustered

Class colour

cluster0 cluster1